



Lecture 37

The Social Implications of Computing

CS61B, Spring 2026 @ UC Berkeley

Josh Hug and Kay Ousterhout

Today, we'll talk about the impact of what you're learning on society.

- We'll start by talking about the jobs of the past (because I don't know what the jobs of the future will be).
- Then we'll talk about the history and future of the world.
- Then your life.

Tech Careers

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Tech Careers

A Brief History of Earth

The Bigger Picture

- How Software Resurfaced Human Society (90s - Present)
- Fulfillment

The mobile game Candy Crush tracks the number of days you have played in a row.

- Every consecutive day gets you a reward.
 - Progress indicator.
 - Up to 2 hours worth of a special item for that day that makes the game more fun.

If you miss a day, the counter resets.

Why does this feature exist in Candy Crush?

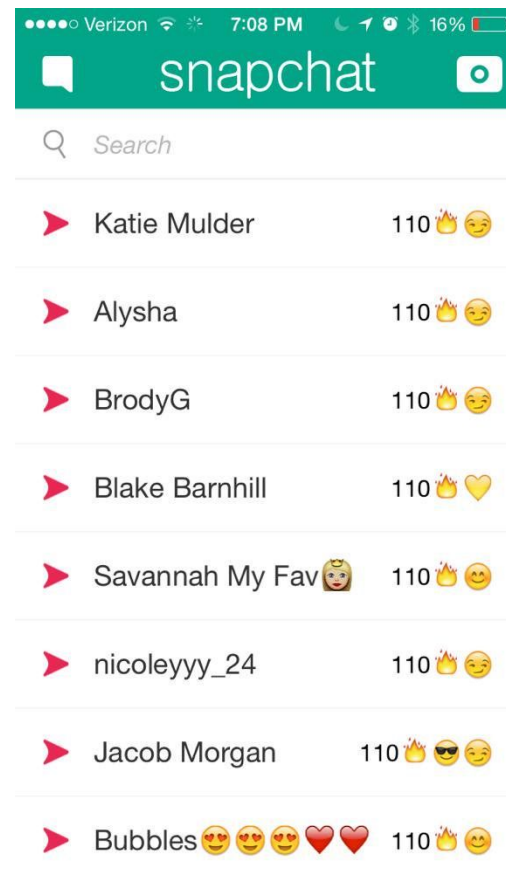
- For addiction, keep you playing
- To increase engagement.
- Reward the player, it's fun.

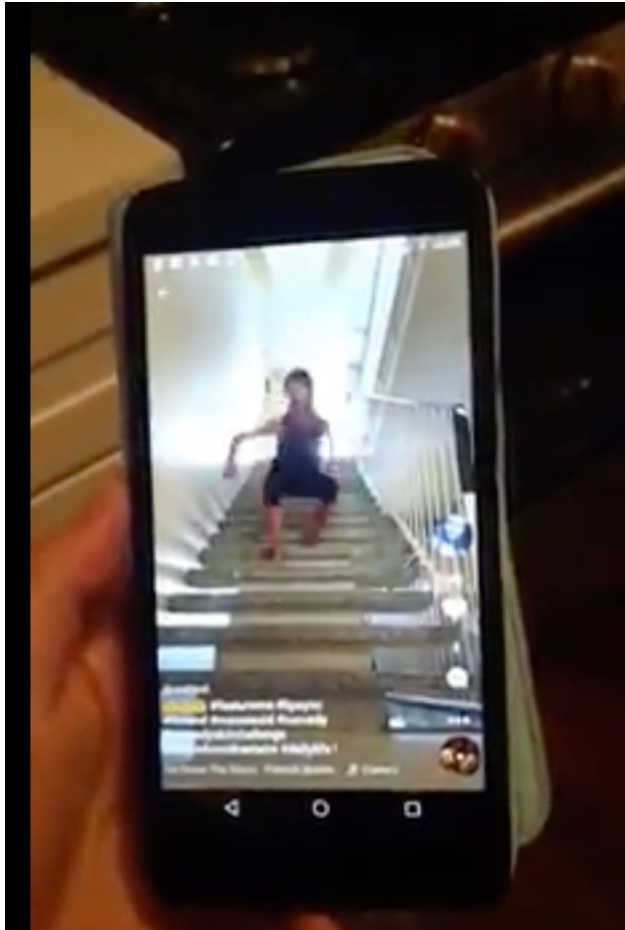


Similarly, for every day that you and a friend communicate on snapchat, your snap streak is extended by one day.

Why does this feature exist in Snapchat?

- Keep people coming back.





Josh Hug
 July 13, 2016 · 🧑

You know how there are 582843 different apps each with their own user base and communication channel? **Sandra Upson** discovered this one. It is... amazing.

This is what you are missing out on by sticking with Facebook, my fellow olds.

See less

6
 5
 296

Like
 Comment
 Share

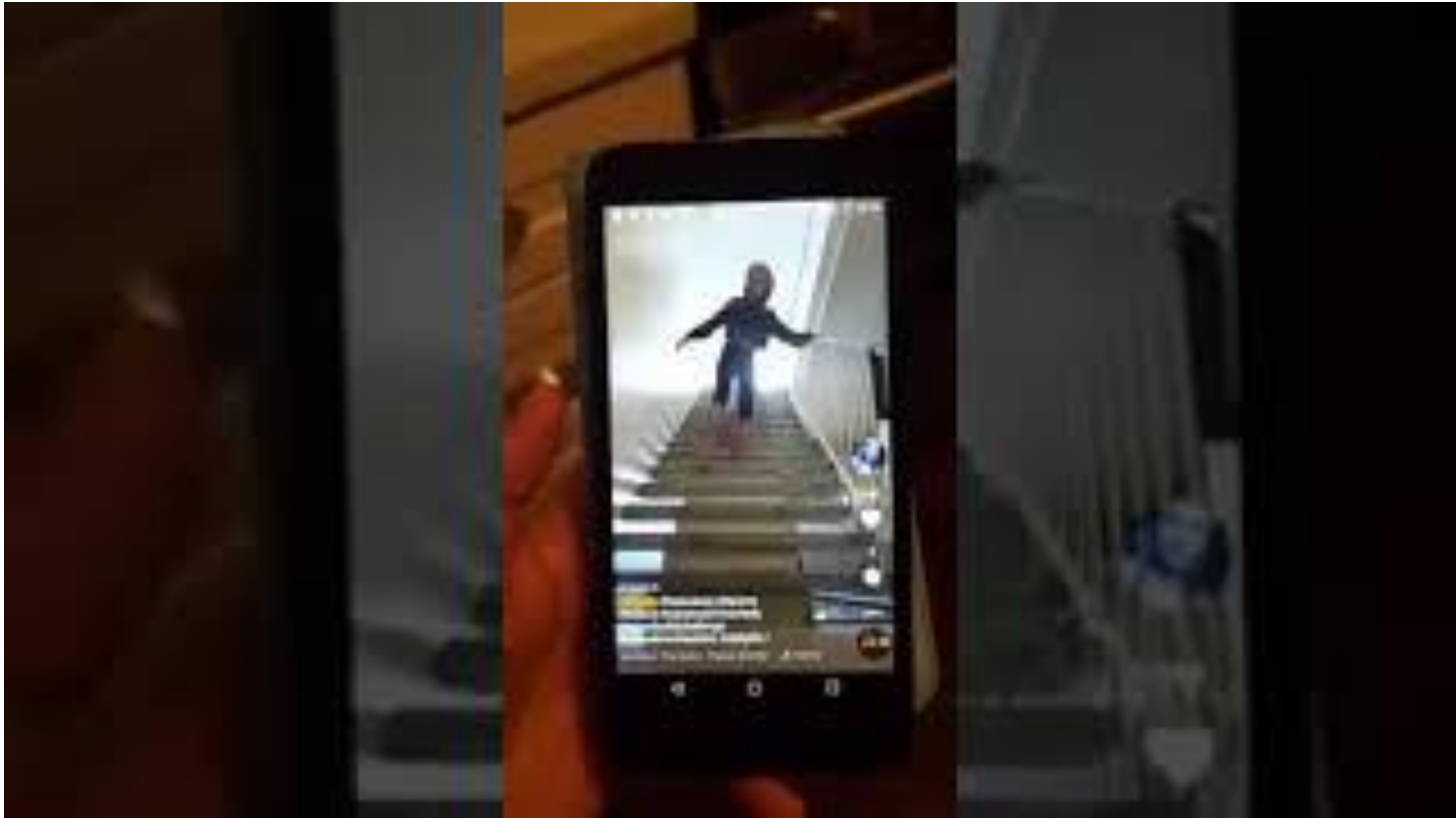
Comments [See all](#)

Josh Hug
 The sound is a critical part of watching this
 9y Like Reply

Audrey Hare
 This is amazing
 9y Like Reply

Reply as Josh Hug

Hussain Gilani
 It's an app where people walk down stairs to music?



The instant you open TikTok, a video is playing.

- It takes milliseconds to swipe to the next video.

Of course, these features exist to increase engagement.



Note: As an elder millennial, I am immune to TikTok's siren song.

Other Engagement Generating Features in Your Life?

- Likes on social media.
- Video games with daily rewards - booth game outside.
- 700 duolingo streak. Ran out of Spanish. Now on to Chess, nature's language.
- Notifications. [I have these off on my phone, let SMS messages in, maybe slack sometimes]
- Autoplay (netflix, amazon video, youtube).
- Oski eats - drives engagement by existing.
 - Calories, menus, it's useful.
-

What positive impacts do these features have on the world?

- Connection and community - can keep up with people.
- “In tools like study apps, they can create a true positive tool build good habits (I have an app that gives me virtual tomatoes to make me happy).”
- Opportunities and learning.



What negative impacts do these features have on the world?

- Addiction and time consumption.
- Shallow interaction -- my online friendships aren't as fruitful.
-

What positive impacts do these features have on the world?



What negative impacts do these features have on the world?



Is TikTok a net positive for the world?

- Overall more net negative, but I will work there for \$150,000/year.

I'm assuming you're familiar with Khan Academy: <https://www.khanacademy.org/>.

- ... but just in case you're not, let's check it out.



**For every student,
every classroom.
Real results.**

We're a nonprofit with the mission to provide a free, world-class education for anyone, anywhere.

Learners

Teachers

Districts

Parents

What positive impacts does Khan Academy have on the world?

- Free education.
- Use it to teach other people, tool for human to human teaching.
- Reduces educational inequality.
- Encourages people to learn by gamifying education.
- Encourage people to make mistake, give guidance when they do.

What negative impacts does Khan Academy have on the world?

- Difficulty to succeed, because answers are very specific. It's a narrow education.
- Some people go overboard, they get hooked, they don't eat. It's chaos.
- Advertises itself as a singular form of learning, doesn't link out to the world.
- Sometimes for students, it can give biased information. If you want to learn history, you're following their curriculum.

Is Khan Academy a net positive?

- Room-wide vote was almost all net positive, a few net-neutral

What are some reasons people might prefer working at TikTok (well, ByteDance) over Khan Academy?

- **Money.** Easier to monetize suffering (addicted customers) over success.
- Brand, more impressive at a party maybe at DNA lounge (death guild on monday).
- **More power of others.**
- Open more career doors later, fancier, more prestige.
- Better networking.
- Down with the CCP. OBSOLETE. Sorry Xi.
- More opportunity for growth.
- Impact.

TikTok vs. Khan Academy

	Revenues	Profits	# Employees	Revenue / Human Employee
ByteDance	\$186 billion (2025)	\$50 billion (2025)	150,000+ (2025)	~\$1,240,000
Khan Academy	\$0.090b (donations) \$0.022b (programs) [2026 figures]	N/A	227 (including 198 humans, 23 dogs, 5 cats, 1 bearded dragon) [April 2025]	~\$60,000 (revenue only) ~\$590,000 (revenue + donations)

Sources: [ByteDance revenue](#), [ByteDance profit](#), [ByteDance employees](#), [Khan Academy Revenue](#), [Khan Academy Employees](#), [Khan Academy Bearded Dragon](#)

A Brief History of Earth

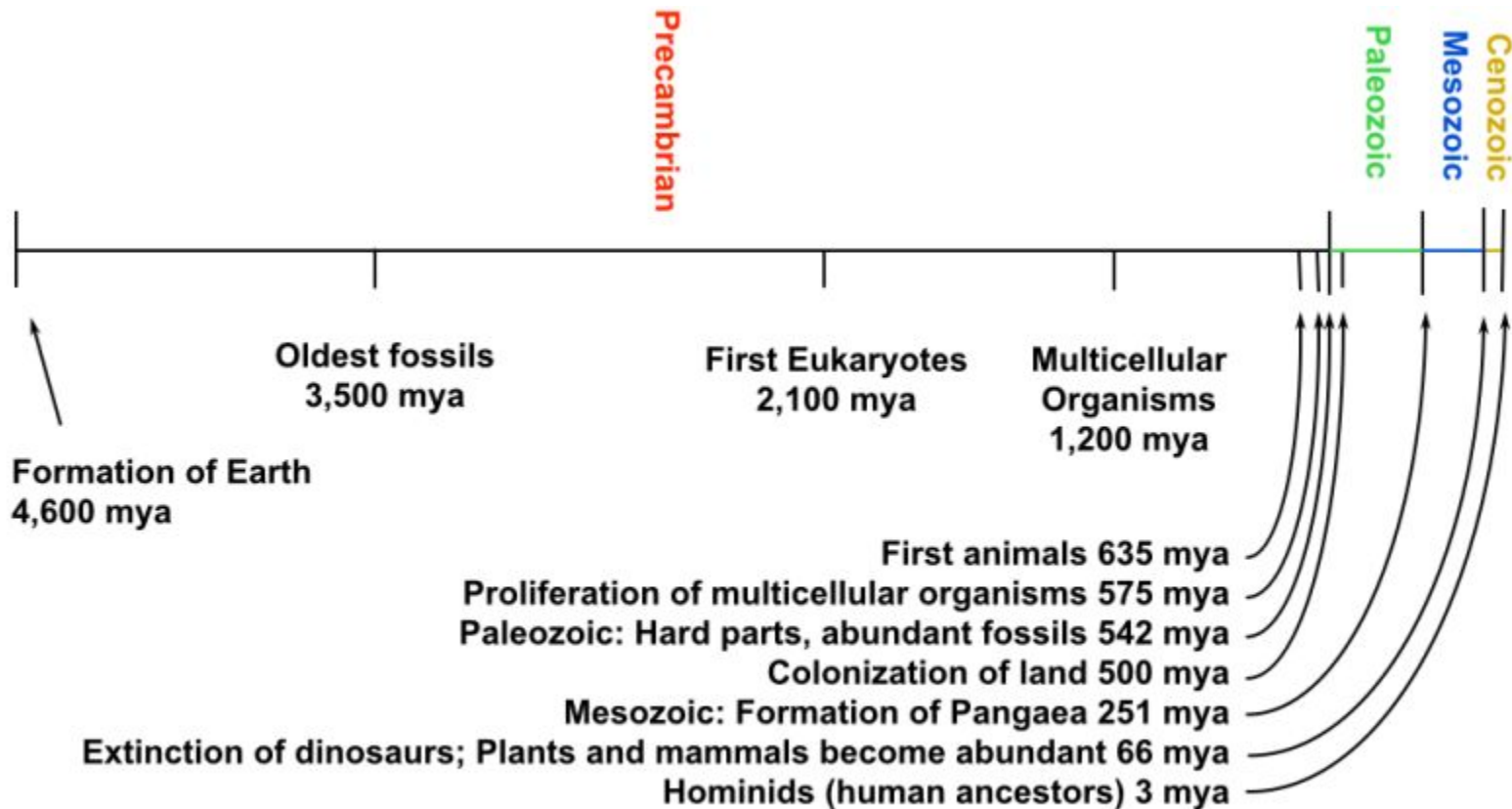
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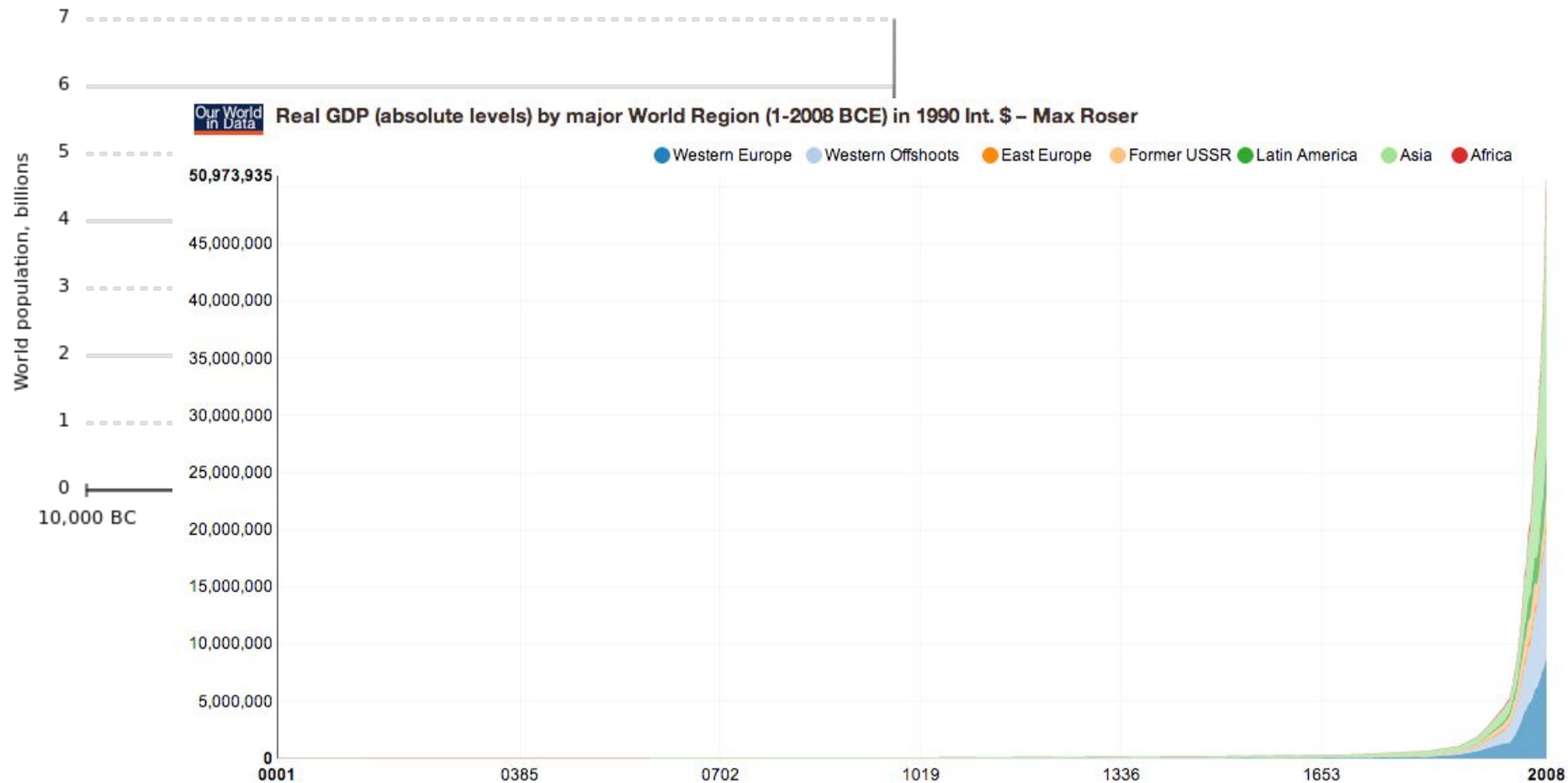
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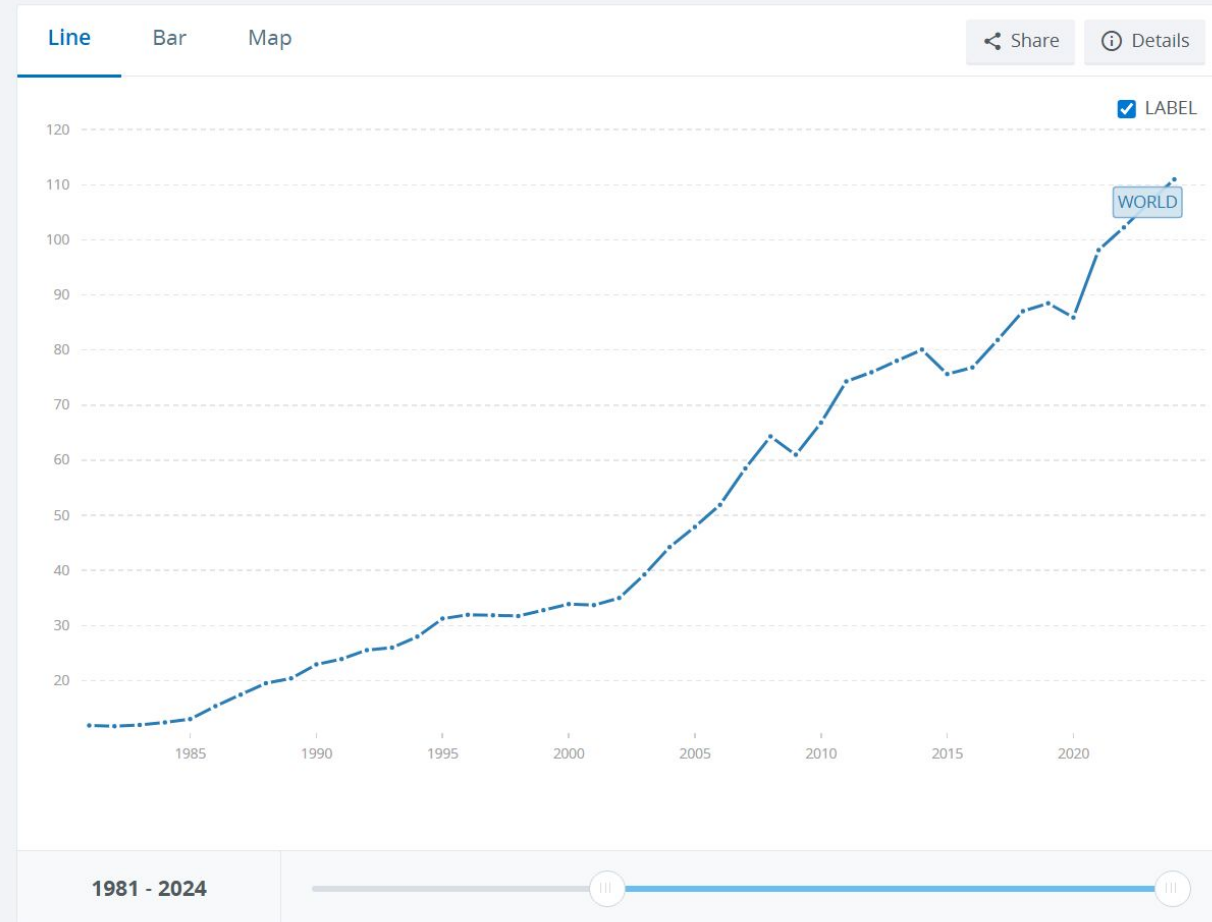


mya = million years ago



Country official statistics, National Statistical Organizations and/or Central Banks; National Accounts data files, Organisation for Economic Co-operation and Development (OECD); Staff estimates, World Bank (WB)

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How Software Resurfaced Human Society (90s - Present)

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A Brief History of Earth

The Bigger Picture

- **How Software Resurfaced Human Society (90s - Present)**
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Programming: purest creative medium in the history of the world?

- Not limited by physical constraints.
- If you can imagine it, you can build it.
- **Limitations are intellectual**: can we understand what we have built?

Quite suddenly, the ceiling on those limitations has been lifted considerably by AI.

- My best guess: The impact on society is going to be **immense**, on par with the discovery of electricity.

Software has completely reshaped society in my lifetime.



1994
Today Show

Today, we'll discuss your role in the future we're creating.



← Thread

 **Sharif Shameem** ✓
@sharifshameem

This is mind blowing.

With GPT-3, I built a layout generator where you just describe any layout you want, and it generates the JSX code for you.

W H A T

Describe a layout.

Just describe any layout you want, and it'll try to render below!

Generate

```
<button style={{backgroundColor: 'pink', border: '2px solid green', borderRadius: '50%', padding: 20, width: 100, height: 100}}>Watermelon</button>
```



1.9M views 0:06 / 2:00

7:01 AM · Jul 13, 2020

<https://twitter.com/sharifshameem/status/1282676454690451457>



hug 5:26 PM

Here are the Fall 2022 final exam grades including the work we did in the gpt61b channel to see how gpt4 did on the exam: https://docs.google.com/spreadsheets/d/1B9LEwseu1j-JjRXCiNQuPXW_MwpqxQqICAYueV-geq0/edit#gid=1659199285

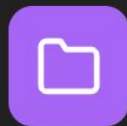
I'd like to give GPT4 a shoutout for earning a score of 0.29 standard deviations above the mean, landing at rank 392 out of the 908 61B students (including ChatGPT)

100

12



GPT o1 was able to earn a score in the top 4% of the class.



p3.zip

Zip Archive

Hi, I've uploaded the starter code for a project in my class and the specification, all in one zip file. Could you fill in the code and give a demo solution including ambition points?

By 2025, ChatGPT o3 could do the entire project from 61B without trouble.

Thought for 2m 37s >

Here's a complete reference implementation together with one ambition feature ("toggle-able line-of-sight lighting").

Download it, unzip, and drop it back into the same repo you gave me (it already preserves the original directory layout):

[Download the solution archive](#)

In 2026, I've found myself using LLMs for things that would have been impossible six months ago.

- Example: Training an LLM to review design docs for project 4B.
- <https://docs.google.com/document/d/1WdQK530GIERcSds-F3Qpz0tGTCInnrYZQHQsKBtRfbl/edit?tab=t.0>

Can you incorporate the lessons from these comments and suggestions into that guide? I'll note here also that it is important that the beginning of every report start with "This report was prepared by Claude Opus 4.6. In past semesters, we did not provide personalized feedback. However, earlier today, I (Josh) was thinking 'I could probably build a pretty good LLM feedback tool.' After many hours of exploration and tuning and guidance, I managed to get it working fairly well. If you have any questions, feel free to post on Ed and a member of the course staff will get back to you. Sorry if the AI is glazing you, I think I got it to mostly stop." Also, the note earlier about always ending reports with a summary is a good one. ■

I hope you are well. I saw on the week 10 survey that an LLM gave feedback on our design docs ("**Project 4B Design Doc LLM Feedback**"). I can't find the feedback on my Gradescope submission nor have I received anything through email. Project 4C is due tonight. Where is the feedback?

P.S. If an LLM is providing me feedback, why are my parents paying tuition?

As for why an LLM: I guess my messaging on the ed post failed but using an LLM here is not a substitute for some higher quality human feedback. In the past, we have not graded designs because it is not remotely feasible to do so. Going through 1100 designs at this level of detail would take ~30 minutes per design or 550 hours of staff time. At \$70/hour for each TA hour, it would cost us around \$38,500 to provide feedback on all designs, which is 10% of the staff budget (not including me) for the course.

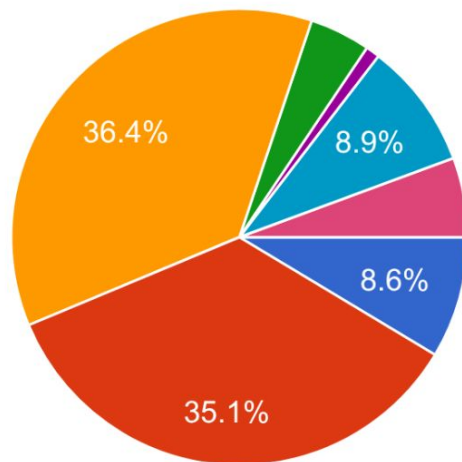
This semester, I decided on a lark to spend a half day of time and \$100 of my own money, and I think I was able to generate feedback that is at the level of human quality. But this was just an experiment. We'll be surveying students to see what they think and if students don't like it, we will stop.

How did you find the LLM-generated WordNet Design Doc Feedback?



How did you find the feedback on your Wordnet design doc?

302 responses



- Very helpful
- Somewhat helpful
- Neutral, e.g., parts were helpful but parts were not
- Somewhat unhelpful or misleading
- Very unhelpful
- I didn't receive the feedback
- I didn't read the feedback



C Fletch 8:52 PM

Not to sound hyperbolic, but from my experience working with Claude for the past month... well, it's the most extraordinary level shift in technology I think I have ever seen (and this is relative to just the SotA last year). I honestly feel like we recently passed the event horizon and everything is *very very* soon going to change. I'm sure folks saw the open Knuth problem that Claude just solved. This is just the beginning.

<https://knightcolumbia.org/content/ai-as-normal-technology>

- Great thoughts on adoption speed.

Clearly, LLM tools are moving fast.

- Getting much better at all sorts of human tasks.
- These tools will (IMO) continue to amplify the power of folks who have the knowledge to use the tools.

Programming: purest creative medium in the history of the world?

- Not limited by physical constraints.
- If you can imagine it, you can build it.
- **Limitations are intellectual**: can we understand what we have built?

Quite suddenly, the ceiling on those limitations has been lifted considerably by AI.

- My best guess: The impact on society is going to be **immense**, on par with the discovery of electricity.

Next Monday, we'll continue this conversation with former 61B TA Stella Kaval ('25).



Stella Kaval 12:26 PM

Hi Josh! Wondering if there's a spot to come back for a guest lecture again? Happy to share more industry insights - especially since we are now 80-100% AI generated code in my org! Brings up lots of new skills like reviewing code and planning before prompting

Fulfillment

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As noted earlier, software is largely unconstrained by the laws of physics.

- Limitation is what you can keep in your head.
- Higher level abstractions yield bigger systems.

Until recently, huge numbers of programmers were needed to translate ideas into code.

- Now, the skill set is growing more advanced.

Historically, STEM experts have been a rare commodity.

Revenue per Employee

Meta	\$2.55 million
Alphabet (Google)	\$2.11 million
Amazon	\$459,000
Microsoft	\$1.24 million
Apple	\$2.51 million

Meta	\$767,000
Alphabet (Google)	\$693,000
Amazon	\$50,000
Microsoft	\$447,000
Apple	\$675,000

The skills you are building will be in high demand from companies, non-profits, government agencies, educational institutions, and more.

- The choice of how to spend your career is yours.

Sources:

<https://investor.atmeta.com/investor-news/press-release-details/2026/Meta-Reports-Fourth-Quarter-and-Full-Year-2025-Results/default.aspx>
- <https://www.sec.gov/Archives/edgar/data/1652044/000165204426000012/googexhibit991q42025.htm>
- <https://www.macrotrends.net/stocks/charts/AMZN/amazon/revenue>
- <https://www.sec.gov/Archives/edgar/data/789019/0000095017025100235/msft-20250630.htm>
- <https://www.mactech.com/2025/11/19/apple-has-164000-employees-that-generate-an-annual-average-of-2413171-each/>

This is a game I played as a kid.



It was pretty trivial to edit your saved game so that your starting town was already huge and prosperous.

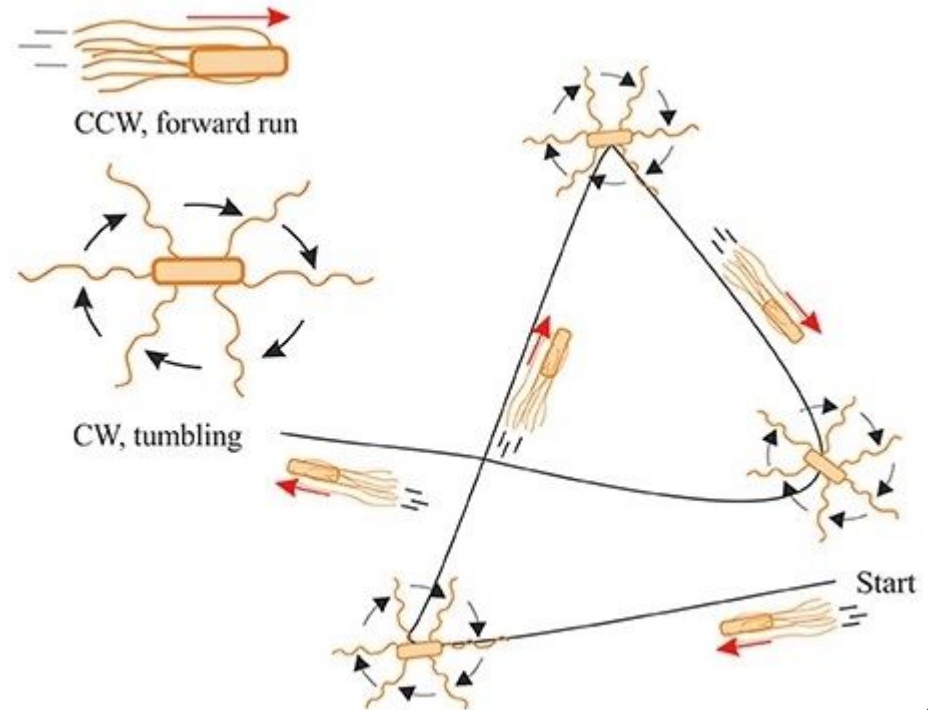
- But the game wasn't fun to play.

IMO, constraints and (achievable, reasonable) struggle make life worth living.

Run-and-Tumble

Studying microbiology in grad school I learned about “[Run and Tumble](#)”.

- Flagella-bearing bacteria use this strategy to climb nutrient **gradients**.



Humans climb gradients as well, gaining happiness in the process.

- Some happiness is more transient (pay increases, a new car).
- And some more durable (relationships, good habits, fulfilling work).

